The many-electron Hamiltonian of NO_2

Write down the many-electron Hamiltonian for NO_2. How many terms are in this Hamiltonian?

$$\hat{H} = \underbrace{\sum_{\alpha} \frac{-\hbar^2}{2M_\alpha}\nabla_\alpha^2}_{\hat{T}_{Nuc}} + \underbrace{\sum_{i} \frac{-\hbar^2}{2m_e}\nabla_i^2}_{\hat{T}_{elec}} + \underbrace{\sum_{\alpha>\beta} \frac{1}{4\pi\epsilon_0}\frac{Z_\alpha e \cdot Z_\beta e}{|\vec{R_\alpha} - \vec{R_\beta}|} - \sum_{\alpha i} \frac{1}{4\pi\epsilon_0}\frac{Z_\alpha e \cdot e}{|\vec{R_\alpha} - \vec{r_i}|} + \sum_{i>j} \frac{1}{4\pi\epsilon_0}\frac{e \cdot e}{|\vec{r_i} - \vec{r_j}|}}_{\hat{V}(\vec{r};\vec{R})}$$

NO_2 : $\left.\begin{array}{l} Z_N = 7 \\ Z_O = 8 \end{array}\right\}$ $7 + 2 \cdot 8 = 23\ e^-$

$$\hat{T}_{nuc} = -\frac{\hbar^2}{2}\left(\frac{\nabla_N^2}{M_N} + \frac{\nabla_{O_1}^2}{M_{O_1}} + \frac{\nabla_{O_2}^2}{M_{O_2}}\right)$$ 3 terms

$$\hat{T}_{elec} = \sum_{i=1}^{23} -\frac{\hbar^2}{2m_e}\nabla_i^2$$ 23 terms

$$\hat{V}_{NN} = \frac{1}{4\pi\varepsilon_0}\left(\frac{7e \cdot 8e}{|\vec{R}_N - \vec{R}_{O_1}|} + \frac{7e \cdot 8e}{|\vec{R}_N - \vec{R}_{O_2}|} + \frac{8e \cdot 8e}{|\vec{R}_{O_1} - \vec{R}_{O_2}|}\right)$$ 3 terms

$$\hat{V}_{EN} = \frac{1}{4\pi\varepsilon_0}\left(\sum_{i=1}^{23} \frac{7e^2}{|\vec{R}_N - \vec{r}_i|} + \sum_{i=1}^{23} \frac{8e^2}{|\vec{R}_{O_1} - \vec{r}_i|} + \sum_{i=1}^{23} \frac{8e^2}{|\vec{R}_{O_1} - \vec{r}_i|}\right)$$ 69 terms

$$\hat{V}_{EE} = \sum_{\substack{i=1 \\ i \neq j}}^{23} \sum_{i>j} \frac{1}{4\pi\varepsilon_0}\frac{e \cdot e}{|\vec{r}_i - \vec{r}_j|}$$

$23 \cdot 22 = 506$

$\frac{506}{2} = 253$ terms

$ij - ji$

351 terms